Movie Recommendation System using Matrix Factorization

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Abstract

We propose to build a movie recommendation system using biased matrix factorization on the MovieLens and Netflix datasets. The system addresses the challenge of sparse user-item interactions by learning latent representations of users and items. We will evaluate performance using RMSE, MAE, Precision@K, and Recall@K, and compare our approach with multiple baselines and advanced models including SVD++ and kNN collaborative filtering. We also plan to explore a simple hybrid extension using movie genres and release year as content features.

1 Introduction

Recommender systems play a crucial role in modern platforms such as Netflix by enabling personalized content delivery. A key challenge in such systems is the sparsity of user-item interaction data. Matrix factorization methods address this by learning latent factors representing user preferences and item characteristics.

2 Related Work

Collaborative filtering has been widely studied in recommendation systems. Matrix factorization techniques gained prominence during the Netflix Prize competition [1]. Extensions such as SVD++ incorporate implicit feedback [2]. Neighborhood-based methods like kNN remain strong baselines due to their simplicity and interpretability.

3 Dataset

We will use both the MovieLens dataset [3] and the Netflix dataset [4] for our experiments. These datasets provide user ratings for movies and are widely used benchmarks in recommendation system research.

The MovieLens dataset includes additional metadata such as genre tags and release year, which enables us to incorporate simple content-based features alongside collaborative filtering. The Netflix dataset, being larger and more sparse, allows us to evaluate the performance and scalability of our models under more challenging conditions.

Using both datasets enables us to study model behavior across different data characteristics and provides a more comprehensive evaluation of recommendation performance. We will split each dataset into training and test sets, and use a validation subset during training for hyperparameter tuning.

4 Objectives

- Build a recommendation system using matrix factorization
- Compare against baseline and advanced models
- Evaluate using rating and ranking metrics
- Analyze performance across user and item segments

5 Methodology

We adopt a collaborative filtering approach using matrix factorization, where latent factors are learned for users and items from observed ratings.

- **Baselines:** global mean, user mean, item mean, and bias-only model
- **Main Model:** biased matrix factorization trained using stochastic gradient descent (SGD)
- **Extensions:**
  - SVD++ to incorporate implicit feedback
  - kNN-based collaborative filtering as a neighborhood-based comparison
  - A simple hybrid extension using content features such as movie genres and release year (available in MovieLens)

We will evaluate these approaches across both MovieLens and Netflix datasets to compare their performance under different levels of sparsity and scale.

6 Evaluation

We will evaluate our models using both rating prediction and ranking-based metrics to capture different aspects of recommendation quality.

Rating Metrics:

- RMSE
- MAE

Ranking-based metrics evaluate the quality of top-K recommendations, taking into account both relevance and ranking order.
Ranking Metrics:

- Precision@K
- Recall@K
- NDCG@K (Normalized Discounted Cumulative Gain)

7 Experiments

We will conduct experiments to study:

- Effect of latent dimension on model performance
- Effect of regularization on overfitting
- Performance across different user activity levels
- Performance across popular vs less popular items
- Comparison of model behavior across MovieLens and Netflix datasets

8 Expected Outcomes

We expect latent factor models to outperform simple baselines in handling sparse rating data. We also expect that incorporating simple content features such as genres and release year may provide improvements, particularly for items with limited interaction data. Additionally, we aim to analyze how model performance varies across datasets with different characteristics, such as sparsity and scale, and understand the strengths and limitations of different approaches.

9 Conclusion

This project aims to build a robust and well-evaluated recommendation system with strong analytical insights.

References

[1] Yehuda Koren, Robert Bell, and Chris Volinsky. 2009. Matrix factorization techniques for recommender systems.
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[4] Javia, Rishit. 2021. Netflix Movie Rating Dataset.